Chirag Agarwal

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Location: Seattle, WA, USA

Innovative Software Engineer specializing in Agentic AI, GenAI, and distributed systems. Founding member of AWS Quick Research and patent holder, passionate about conversational AI architectures and building scalable global platforms.

Technical Skills: Java, Python, React, C++, JavaScript, TypeScript, PHP, SQL, C#, .NET

Education

- **Viterbi School of Engineering, University of Southern California** — Los Angeles, CA, USA
  Masters of Science in Computer Science, Current CGPA: ***3.9/4.0*** — *January 2019 – December 2020*
- **Manipal Institute of Technology** — Manipal, KA, India
  Bachelor of Technology in Computer Science and Engineering, CGPA: ***8.81/10.0*** — *July 2011 – June 2015*

Work Experience

- **Amazon Web Services (AWS), Agentic AI** — Seattle, WA, USA
  Software Development Engineer 2 — *July 2025 – Present*
  - **Founding Member - Quick Research**: Served as a founding member of **Quick Research** (AWS Agentic AI), a service for autonomous deep research. Designed and implemented the **Retrieval-Augmented Generation (RAG)** architecture that enables users to securely integrate internal organizational data (via Quick Suite) to contextually ground and guide the agent's research.
  - **Third-Party Data Integrations**: Engineered secure tool-use integrations for premium third-party financial data providers (e.g., **FactSet, S&P**), empowering the research agent to autonomously query, retrieve, and synthesize external market data alongside internal user context.
- **Amazon.com, Inc.** — Seattle, WA, USA
  Software Development Engineer 2, Software Development Engineer 1 — *February 2021 – July 2025*
  - **Customer Service Chat Initiation**: Implemented a high-availability REST API for initiating customer service chats, integrating with six critical backend systems to manage customer and chat records. The API now handles **100% of global Amazon customer service chat traffic**, supporting millions of daily interactions with high scalability.
  - **Migration to GenAI-Based Customer Service Chatbot**: Contributed to migrating Amazon's legacy Customer Service Chatbot to a GenAI-powered platform by integrating state-of-the-art **LLM models** for customer intent detection and resolutions. Expanded the chatbot's accessibility across all customer device types, improving coverage and user engagement.
  - **Self-service Concessions Resolutions**: Led the integration of AI-powered chat and voice bots with multiple systems to automate resolutions for high volume use cases such as 'item not received' and 'missing shipment items'. This platform enhancement resulted in a **1.5 Million** annual agent contact reduction in the US, with ongoing worldwide expansion.
  - **LLM-Based Customer Satisfaction Analysis**: Led the ideation and implementation of a silent customer satisfaction scoring system using Large Language Models (LLM) for customer service chats, analyzing transcripts to predict satisfaction scores with high accuracy while comparing against actual customer feedback, resulting in improved service quality measurement without increasing customer survey fatigue.
  - **Legacy Service Deprecation and GraphQL Migration**: Led the deprecation of a legacy service responsible for fetching and processing customer data for chat bot resolutions. Designed and executed a migration plan to expose APIs via a reusable **GraphQL** interface, enabling standardized data access throughout the organization. Improved operational excellence by over **20%** through reduced maintenance overhead and enhanced platform efficiency.
  - **Interactive Reusable Widgets**: Developed and deployed interactive, reusable widgets across multiple customer service platforms for tracking orders and refunds, streamlining user experience and ensuring consistency across the organization.
  - **Security Risk Mitigation**: Designed and implemented a robust solution to address critical vulnerabilities across all Customer Service bot platforms, effectively preventing exploitation by malicious actors and enhancing platform security.
  - **Customer Service Worldwide Hackathon Winner**: Won the 'Judge's Favorite' award twice for independently conceived and executed solutions:
    - **[2024] GenAI-Powered Code Documentation**: Developed a platform using GenAI models on AWS Bedrock to auto-generate and update code documentation, streamlining and boosting developer productivity across Amazon.
    - **[2022] Cross-Platform Chat Notification System**: Developed a cross-platform mobile notification system, enabling real-time, interactive updates for customers. This solution significantly reduced chat park rates and enhanced user experience by allowing customers to reply and engage from any device.
- **Amazon.com, Inc.** — Seattle, WA, USA
  Software Development Engineer Intern — *May 2020 – August 2020*
  - **Multi-Item Concessions Support**: Enabled the Customer Service Chatbot to handle concessions for multi-item Amazon shipments by developing backend support for various issue scenarios. Additionally, built a multi-item picker using **React** to allow customers to select items for concessions, reducing agent contacts by an over **100,000** annually in the US.
- **USC Norris Comprehensive Cancer Center** — Los Angeles, CA, USA
  Student Software Developer — *May 2019 – May 2020*

- **Publications Portal**: Developed a cancer research publications platform used by 8 leading cancer centers across the US, including Stanford and Norris, to streamline access and collaboration on oncology research.

- **Citrix Systems, Inc.** Bangalore, KA, India
  Software Engineer 2, Software Engineer 1, Software Engineering Intern *January 2015 – November 2018*
  - **Product Usage Analytics**: Built an end-to-end analytics pipeline using **Python** and **Tableau** to analyze customer's product usage data from sources like Citrix's CEIP database, Google Analytics, and App stores. Enabled data-driven business and product decisions across all Citrix products.
  - **Citrix Video Library**: Led a team of 4 and collaborated with 3 cross-functional teams to deploy the internal video library.
  - **Customer Case Analytics**: Designed a **Python**-based framework to fetch, filter, and analyze Salesforce support cases. Applied text classification for case categorization, improving triage and resolution efficiency.
  - **Citrix Translations Platform**: Developed the internal translations platform and exposed **REST** services using **.NET** and **C#** to manage UI string translations with real-time status tracking, aiding teams in monitoring translation progress.
  - **Citrix Documentation Feedback**: Built the feedback collection and analytics system for docs.citrix.com, improving content insights and user experience.
  - **Voice of Customer Analysis**: Developed a sentiment analytics system to automate data retrieval from App stores and Twitter. Leveraged **NLTK** with **Python** for textual review analysis, aiding key business decisions.
  - **Lean Innovation Challenge**: Recognized twice in Citrix's yearly Lean Innovation Challenges for pioneering ideas:
    - **[2018] Workflow Automation**: Ranked top 3 out of 98 submissions and founded an internal startup. Proposed a workflow automation product to enhance employee productivity.
    - **[2017] Secure Offline File Sharing**: Ranked top 5 out of 150 submissions. Designed and patented a secure enterprise offline file-sharing solution.
  - **Automation Results Manager**: Developed a platform to track and analyze the globalization automation testing results across all Citrix products. Built log parsers for diverse formats and a backend service for efficient database interactions.

Projects

- **Brain-Computer Interface for IoT Control**: Developed a BCI prototype to enable hands-free control of IoT devices and computers, improving accessibility for individuals with paralysis. Focused on signal processing and real-time system integration.
- **Weather applications**: Built a real-time web app using **Angular** and **Node.js** along with an **Android** app for dynamic weather tracking, ensuring high availability and efficient data fetching – CSCI-571 course project.
- **Recommender systems**: Designed and implemented Model, User, and Item-based collaborative filtering recommendation engines on the Yelp dataset using **Spark** with **Python** and **Scala**, optimizing for performance and scalability – INF-553 course project.
- **Stock Market Prediction with Big Data**: Developed a sentiment-driven stock price prediction model for oil securities using **MapReduce** on large-scale unstructured data from Twitter and news feeds, leveraging distributed computing for high-speed analytics.

Achievements and Awards

- **F8 2019 Hackathon**: Invited by Facebook to the prestigious F8 hackathon and conference. Developed an **Augmented Reality**-based education platform using **Spark AR Studio** and **React**, and presented it to Meta CEO **Mark Zuckerberg**.
- **HackSC**: Won the Facebook Award for developing BSMART, an innovative **AR-powered IoT** platform to control smart devices and Tesla cars.
- **Citrix Spot Awards**: Received two Spot Awards for impactful contributions with 'Product Usage Analytics' and 'Citrix Video Library'.
- **Citrix Quality Excellence Awards**: Recognized with three Quality Awards for driving improvements with 'Customer Case Analytics', 'Product Usage Analytics', and the 'Citrix Translations Platform'.

Patents

- **Authentication for secure file sharing**: Patent No. US 11,831,646 B2, Issued Nov 2023
- **Secure offline streaming of content**: Patent No. US 11,010,486 B2, Issued May 2021
- **Cloud authenticated offline file sharing**: Patent No. US 10,834,170 B2, Issued Nov 2020